

GRouNdGAN Generator Parameter-Count Scaling

Analytical scaling analysis

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1 Purpose

This report derives how the causal-generator parameter count changes when the number of transcription factors, target genes, top- k regulators, depth, width, or noise variables is varied. It also gives the equivalent expression in terms of the number of imposed edges.

The analysis describes the causal generator implemented in `src/networks/masked_causal_generator.py`. It assumes that every target gene has exactly k selected regulators, so that the number of imposed edges is

$$E = kG, \tag{1}$$

where G is the number of target genes.

2 Definitions and Assumptions

Table 1: Symbols used in the scaling equations.

Symbol	Meaning
F	Number of transcription factors
G	Number of target genes
$N = F + G$	Total genes, assuming TFs and targets are disjoint
k	Top- k regulators selected per target gene
$E = kG$	Number of imposed edges
n	Noise variables per target gene
w	Width scale per gene
d	Depth per gene
$q = k + n$	Regulators plus noise variables per target
$h = wq$	Hidden width of one target-gene block

The derivation counts trainable weights, biases, and BatchNorm affine parameters. BatchNorm running statistics and implementation-specific state-dict buffers can add small linear terms to the values reported by checkpoint-inspection tools.

3 Causal-Graph Generator

For target gene i , let r_i be its number of regulators. Under the fixed-top- k assumption, $r_i = k$ and every target block has width $h = w(k + n)$.

The sparse generator contains three types of masked linear layers:

$$P_{\text{input}} = G w(k + n)^2, \quad (2)$$

$$P_{\text{hidden}} = dG w^2(k + n)^2, \quad (3)$$

$$P_{\text{output}} = Gw(k + n). \quad (4)$$

Including SparseLinear biases and BatchNorm affine parameters gives the causal-graph-specific generator count

$$P_{\text{graph}}(F, G, k, d, w, n) = G \left[w(k + n)^2 + dw^2(k + n)^2 + (3d + 4)w(k + n) + 1 \right]. \quad (5)$$

The TF count F does not appear in this term because SparseLinear stores parameters only for the selected regulator connections. F controls the available TF feature space and constrains k , but not the number of selected connections when k is fixed.

4 Causal-Controller Contribution

The causal controller is a dense MLP whose final layer outputs all $N = F + G$ genes. Let its latent dimension be z , and let its hidden widths be $a_1, \dots, a_m$, with $a_0 = z$. The trainable controller parameter count is

$$P_{\text{CC}}(F, G) = \sum_{j=1}^m [(a_{j-1} + 1)a_j + 2a_j] + (a_m + 1)(F + G). \quad (6)$$

The full generator count, when the pretrained causal controller is included in the generator state dictionary, is therefore

$$P_{\text{gen}} = P_{\text{CC}}(F, G) + P_{\text{graph}}(F, G, k, d, w, n). \quad (7)$$

If the controller is counted separately, omit P_{CC} .

5 Partial Scaling Functions

Each expression below varies one variable while all other variables remain fixed.

5.1 Transcription Factors

$$P(F) = C_0 + L(F + G) + K \quad (8)$$

where $L = a_m + 1$, C_0 contains controller terms independent of F and G , and K is the causal-graph contribution. Thus, parameter count is linear in F .

5.2 Target Genes

$$P(G) = C_0 + LF + G \left[L + w(k + n)^2 + dw^2(k + n)^2 + (3d + 4)w(k + n) + 1 \right]. \quad (9)$$

With k, d, w, n fixed, parameter count is linear in G .

5.3 Top- k

$$P(k) = C_0 + L(F + G) + G \left[(w + dw^2)(k + n)^2 + (3d + 4)w(k + n) + 1 \right]. \quad (10)$$

Therefore, parameter count is quadratic in k .

5.4 Depth

$$P(d) = C_0 + L(F + G) + G \left[w(k + n)^2 + d(w^2(k + n)^2 + 3w(k + n)) + 4w(k + n) + 1 \right]. \quad (11)$$

Parameter count is linear in d .

5.5 Width

$$P(w) = C_0 + L(F + G) + G \left[d(k + n)^2 w^2 + ((k + n)^2 + (3d + 4)(k + n))w + 1 \right]. \quad (12)$$

Parameter count is quadratic in w because the fully connected hidden blocks scale with the square of each block's width.

5.6 Noise Per Gene

$$P(n) = C_0 + L(F + G) + G \left[(w + dw^2)(k + n)^2 + (3d + 4)w(k + n) + 1 \right]. \quad (13)$$

Parameter count is quadratic in n .

5.7 Imposed Edges

Using $k = E/G$ gives

$$P(E) = C_0 + L(F + G) + G \left[(w + dw^2) \left(\frac{E}{G} + n \right)^2 + (3d + 4)w \left(\frac{E}{G} + n \right) + 1 \right]. \quad (14)$$

For fixed F , G , d , w , and n :

$$P(E) = AE^2 + BE + C. \quad (15)$$

The leading coefficient is proportional to $(w + dw^2)/G$. This is the analytical reason that the top- k experiment exhibits quadratic generator-parameter scaling with imposed edges.

6 Interpretation

The quadratic term is caused by the block-diagonal, fully connected hidden masks inside each target-gene generator. For target i , the dominant contribution is proportional to $(r_i + n)^2$. Under uniform top- k selection, $r_i = k$ and $E = kG$, yielding the quadratic dependence on E .

For arbitrary causal graphs, total edge count alone is not sufficient. The more general scaling is

$$P_{\text{graph}} \propto \sum_{i=1}^G (r_i + n)^2. \quad (16)$$

Two graphs with the same E can therefore have different parameter counts if their regulator degrees are distributed differently. A highly concentrated regulator distribution produces a larger sum of squares than a uniform distribution.

7 Implementation References

The derivation corresponds to the following implementation details:

- Per-gene hidden width: `masked_causal_generator.py`, lines 208–212.
- Input and noise connections: lines 214–224.
- Fully connected hidden blocks: lines 226–230.
- Per-gene output connections: lines 232–234.
- Repetition over `depth_per_gene`: lines 264–272.
- Sparse connectivity construction: lines 305–316.